

Optimization problems

Let $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ ($0 \leq i \leq m$). Generic **nonlinear program**

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & x \in \{\text{dom } f_0 \cap \text{dom } f_1 \cdots \cap \text{dom } f_m\}. \end{aligned}$$

Henceforth, we drop condition on domains for brevity.

- If f_i are **differentiable** — smooth optimization
- If any of the f_i is **non-differentiable** — nonsmooth optimization
- If all f_i are **convex** — convex optimization
- If $m = 0$, i.e., only f_0 is there — **unconstrained** minimization

Example: Linear programming, Basic ideas of the simplexe method.

Convex optimization problems

Standard form

$$\begin{array}{ll}\min & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b.\end{array}$$

Some observations

- ▶ All f_i are convex
- ▶ Direction of inequality $f_i(x) \leq 0$ **crucial**
- ▶ The only equality constraints we allow are affine
- ▶ This ensures, set of feasible solutions is also **convex**

Convex optimization problems

Def. We denote by $\mathcal{X}$ the **feasible set**

$$\mathcal{X} := \{x \in \mathbb{R}^n \mid f_i(x) \leq 0, 1 \leq i \leq m, Ax = b\}.$$

Def. We denote by p^* the **optimal value** of the problem.

$$p^* := \inf \{f_0(x) \mid x \in \mathcal{X}\}$$

- ▶ If $\mathcal{X}$ is empty, we say problem is **infeasible**
- ▶ By **convention**, we set $p^* = +\infty$ for infeasible problems
- ▶ If $p^* = -\infty$, we say problem is **unbounded below**.
- ▶ Example, $\min x$ on $\mathbb{R}$, or $\min -\log x$ on $\mathbb{R}_{++}$
- ▶ Sometimes **minimum doesn't exist** (as $x \rightarrow \pm\infty$)
- ▶ Say $f_0(x) = 0$, problem is called **convex feasibility**

Optimality

Def. A point $x^* \in \mathcal{X}$ is **locally optimal** if $f(x^*) \leq f(x)$ for all x in a **neighborhood** of x^* . **Global** if $f(x^*) \leq f(x)$ for **all** $x \in \mathcal{X}$.

Theorem For convex problems, locally optimal point also global.

- ▶ Let x^* be a local minimizer of $f(x)$ on $\mathcal{X}$
- ▶ Let $y \in \mathcal{X}$ be any other **feasible** point.
- ▶ We need to show that $f(y) \geq f(x^*) = p^*$.
- ▶ If $y \notin \text{dom } f$, then by definition $f(y) = +\infty$; nothing to prove
- ▶ $\mathcal{X}$ is cvx., so we have $x_\theta = \theta y + (1 - \theta)x^* \in \mathcal{X}$ for $\theta \in (0, 1)$
- ▶ Since f is cvx, and $x^*, y \in \text{dom } f$, we have

$$f(x_\theta) - f(x^*) \leq \theta(f(y) - f(x^*)).$$

- ▶ Since x^* is a local minimizer, for small enough $\theta > 0$, lhs ≥ 0 .
- ▶ So rhs is also nonnegative, proving $f(y) \geq f(x^*)$ as desired.

Optimal set

The set of optimal solutions $\mathcal{X}^*$ may be empty

Example If $\mathcal{X} = \emptyset$, i.e., no feasible solutions, then $\mathcal{X}^* = \emptyset$

Example When only inf and not min, e.g., $\inf e^x$ as $x \rightarrow -\infty$
or $\inf 1/x$ as $x \rightarrow \infty$; so sometimes we have to worry about $\mathcal{X}^* = \emptyset$

Exercise: Verify that $\mathcal{X}^*$ is always convex.

First-order optimality conditions

Theorem Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable in an open set S containing x^* , a local minimum of f . Then, $\nabla f(x^*) = 0$.

Proof: Consider function $g(t) = f(x^* + td)$, where $d \in \mathbb{R}^n$; $t > 0$. Since x^* is a local min, for small enough t , $f(x^* + td) \geq f(x^*)$.

$$0 \leq \lim_{t \downarrow 0} \frac{f(x^* + td) - f(x^*)}{t} = \frac{dg(0)}{dt} = \langle \nabla f(x^*), d \rangle.$$

Similarly, using $-d$ it follows that $\langle \nabla f(x^*), d \rangle \leq 0$, so $\langle \nabla f(x^*), d \rangle = 0$ **must hold**. Since d is arbitrary, $\nabla f(x^*) = 0$.

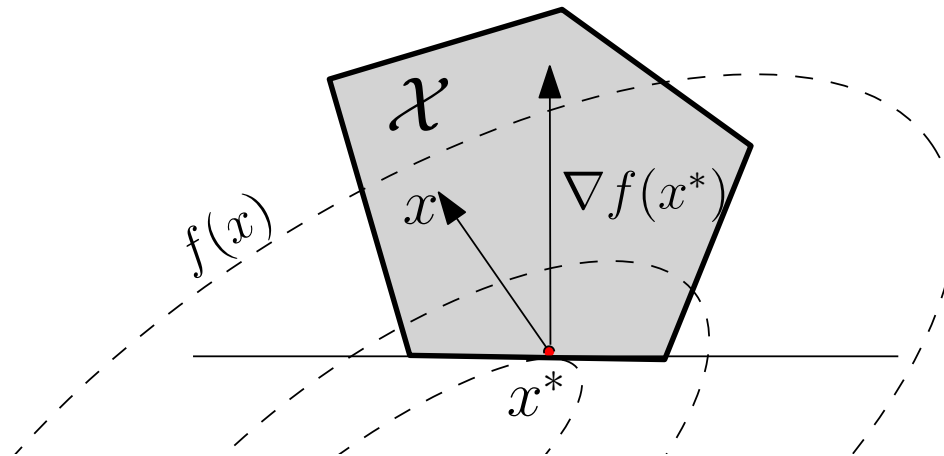
Exercise: Prove that if f is convex, then $\nabla f(x^*) = 0$ is actually **sufficient** for global optimality! For general f this is **not** true. (This property that makes convex optimization special!)

Optimality conditions – constrained

- ♠ For every $x, y \in \text{dom } f$, we have $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$.
- ♠ Thus, x^* is optimal if and only if

$$\langle \nabla f(x^*), y - x^* \rangle \geq 0, \quad \text{for all } y \in \mathcal{X}.$$

- ♠ If $\mathcal{X} = \mathbb{R}^n$, this reduces to $\nabla f(x^*) = 0$



- ♠ If $\nabla f(x^*) \neq 0$, it defines supporting hyperplane to $\mathcal{X}$ at x^*

Optimality conditions – constrained

- Suppose $\exists y \in \mathcal{X}$ such that $\langle \nabla f(x^*), y - x^* \rangle < 0$
- Using mean-value theorem of calculus, $\exists \xi \in [0, 1]$ s.t.

$$f(x^* + t(y - x^*)) = f(x^*) + \langle \nabla f(x^* + \xi t(y - x^*)), t(y - x^*) \rangle$$

(we applied MVT to $g(t) := f(x^* + t(y - x^*))$)

- For sufficiently small t , since ∇f continuous, by assump on y , $\langle \nabla f(x^* + \xi t(y - x^*)), y - x^* \rangle < 0$
- This in turn implies that $f(x^* + t(y - x^*)) < f(x^*)$
- Since $\mathcal{X}$ is convex, $x^* + t(y - x^*) \in \mathcal{X}$ is also feasible
- Contradiction to local optimality of x^*

Optimality – nonsmooth

Theorem (Fermat's rule): Let $f_0 : \mathbb{R}^n \rightarrow (-\infty, +\infty]$. Then,

$$\text{Argmin } f_0 = \text{zer}(\partial f_0) := \{x \in \mathbb{R}^n \mid 0 \in \partial f_0(x)\}.$$

Proof: $x \in \text{Argmin } f_0$ implies that $f_0(x) \leq f_0(y)$ for all $y \in \mathbb{R}^n$.
Equivalently, $f_0(y) \geq f_0(x) + \langle 0, y - x \rangle \quad \forall y, \Leftrightarrow 0 \in \partial f_0(x)$.

Nonsmooth optimality

$$\min \quad f_0(x) \quad \text{s.t. } x \in \mathcal{X}$$

$$\min \quad f_0(x) + \mathbb{I}_{\mathcal{X}}(x).$$

Optimality – nonsmooth

- ▶ Minimizing x must satisfy: $0 \in \partial(f_0 + \mathbb{I}_{\mathcal{X}})(x)$
- ▶ **(CQ)** Assuming $\text{ri}(\text{dom } f_0) \cap \text{ri}(\mathcal{X}) \neq \emptyset$, $0 \in \partial f_0(x) + \partial \mathbb{I}_{\mathcal{X}}(x)$
- ▶ Recall, $g \in \partial \mathbb{I}_{\mathcal{X}}(x)$ iff $\mathbb{I}_{\mathcal{X}}(y) \geq \mathbb{I}_{\mathcal{X}}(x) + \langle g, y - x \rangle$ for all y .
- ▶ So $g \in \partial \mathbb{I}_{\mathcal{X}}(x)$ means $x \in \mathcal{X}$ and $0 \geq \langle g, y - x \rangle \ \forall y \in \mathcal{X}$.
- ▶ **Normal cone:**

$$\mathcal{N}_{\mathcal{X}}(x) := \{g \in \mathbb{R}^n \mid 0 \geq \langle g, y - x \rangle \ \forall y \in \mathcal{X}\}$$

Application. $\min f_0(x) \quad \text{s.t. } x \in \mathcal{X}$:

- ◇ If f_0 is diff., we get $0 \in \nabla f(x^*) + \mathcal{N}_{\mathcal{X}}(x^*)$
- ◇ $-\nabla f(x^*) \in \mathcal{N}_{\mathcal{X}}(x^*) \iff \langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in \mathcal{X}$.

Example

$$\min f(x) \quad \|x\| \leq 1.$$

A point x is optimal if and only if

$$x \in \text{dom } f, \quad \|x\| \leq 1, \forall y \text{ s.t. } \|y\| \leq 1 \implies \nabla f(x)^T (y - x) \geq 0.$$

In other words

$$\begin{aligned} \forall \|y\| \leq 1, \quad \nabla f(x)^T y &\geq \nabla f(x)^T x \\ \forall \|y\| \leq 1, \quad -\nabla f(x)^T y &\leq -\nabla f(x)^T x \\ \sup\{-\nabla f(x)^T y \mid \|y\| \leq 1\} &\leq -\nabla f(x)^T x \\ \|\nabla f(x)\|_* &\leq -\nabla f(x)^T x \\ \|\nabla f(x)\|_* &\leq -\nabla f(x)^T x. \end{aligned}$$

Observe: If constraint not satisfied strictly at optimum ($\|x\| < 1$), then $\nabla f(x) = 0$ (else we'd violate the last inequality above).

Monotonic transformation

Standard form

$$\begin{array}{ll}\min & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b.\end{array}$$

- Say $\psi_0 : \mathbb{R} \rightarrow \mathbb{R}$ is monotone increasing
- $\psi_i : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\psi_i(u) \leq 0$ **iff** $u \leq 0$
- $h(z) = 0$ **iff** $z = 0$.

Transformed problem

$$\begin{array}{ll}\min & \psi_0(f_0(x)) \\ \text{s.t.} & \psi_i(f_i(x)) \leq 0, \quad i = 1, \dots, m \\ & h(Ax - b) = 0.\end{array}$$

Can destroy convexity

Example

$$\begin{aligned} \min \quad & \|Ax - b\| \\ \min \quad & \|Ax - b\|^2 \end{aligned}$$

- ♣ Set of optimal points same
- ♣ Problems equivalent but not same
- ♣ First problem is nondifferentiable
- ♣ Second is differentiable – solvable in closed form!

Epigraph form

Standard form; optimal value p^*

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b. \end{aligned}$$

Epigraph form

$$\begin{aligned} \min_{(x,t)} \quad & t \\ \text{s.t.} \quad & f_0(x) - t \leq 0 \\ & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b. \end{aligned}$$

At the optimum, $t = p^*$.

In other words: Define **sublevel set** $L_t := \{x \mid f_0(x) \leq t\}$, $t \in \mathbb{R}$.

We seek smallest t such that L_t intersects with constraint set $\mathcal{X}$.

Variable elimination

$$\min_{x,y} f_0(x,y) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad i = 1, \dots, m.$$

Recall, since f_0 is convex in (x, y) , $\inf_y f_0(x, y)$ is still convex.

Variable elimination

$$\min_x \tilde{f}_0(x) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad i = 1, \dots, m$$

$$\text{where } \tilde{f}_0(x) := \inf_y f_0(x, y).$$

More generally: $\tilde{f}_0(x) := \inf \{f_0(x, y) \mid g_i(y) \leq 0, i = 1, \dots, m'\}$

Independent constraints important here.

Equality constraint elimination

Standard form

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b. \end{aligned}$$

- For $Ax = b$ to be feasible, $b \in \mathcal{R}(A)$.
- Let x_0 be any solution to $Ax = b$.
- Let F be a matrix with $\mathcal{R}(F) = \mathcal{N}(A)$; so $AFz = 0$
- General solution to $Ax = b$ is of form: $Fz + x_0$

Elimination form

$$\begin{aligned} \min \quad & f_0(Fz + x_0) \\ \text{s.t.} \quad & f_i(Fz + x_0) \leq 0, \quad 1 \leq i \leq m. \end{aligned}$$

Introducing equality constraints

Separable function

$$\begin{aligned} \min \quad & \sum_{i=1}^T f_{0,i}(\mathbf{x}) \\ \text{s.t.} \quad & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m. \end{aligned}$$

Often useful trick: **variable splitting**

$$\begin{aligned} \min_{\mathbf{x}_1, \dots, \mathbf{x}_T, \mathbf{z}} \quad & \sum_{i=1}^T f_{0,i}(\mathbf{x}_i) \\ \text{s.t.} \quad & f_i(\mathbf{z}) \leq 0, \quad i = 1, \dots, m \\ & \mathbf{x}_i = \mathbf{z}, \quad i = 1, \dots, T. \end{aligned}$$

Almost separate problems! Useful for distributed computing.

Constraint removal

Constrained problem

$$\min f_0(x) \quad \text{s.t.} \quad x \in \mathcal{X}.$$

Unconstrained problem

$$\min f_0(x) + \mathbb{I}_X(x)$$

Penalized form (approximate)

$$\min f_0(x) + \rho \|\max\{0, f(x)\}\|_2^2,$$

where $f(x) = [f_1(x), \dots, f_m(x)]^T$; $\rho \gg 0$.

Reducing number of constraints

$$\begin{aligned} & \min f_0(x) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad i = 1, \dots, m \\ \implies & \min f_0(x) \quad \text{s.t.} \quad [g(x) := \max_{1 \leq i \leq m} f_i(x)] \leq 0. \end{aligned}$$

Implicit constraints

$$\min \quad c^T x - \sum_{i=1}^m \log(b_i - a_i^T x),$$

where $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and a_i^T are rows of $A \in \mathbb{R}^{m \times n}$.

- ▶ Implicit constraint: $b_i - a_i^T x \in \text{dom}(-\log z)$ (else $+\infty$)
- ▶ Thus, x must be in **strict interior** of $\mathcal{P} = \{x \mid Ax \leq b\}$.

Idea comes up again in interior point methods
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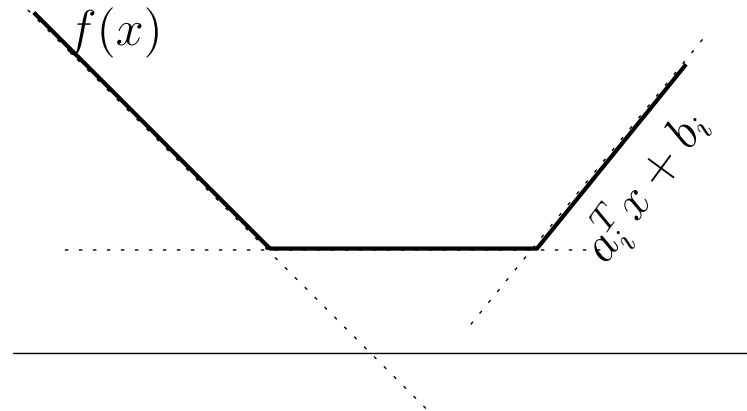
Problem classes

Linear Programming

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b, \quad Cx = d.\end{array}$$

Piecewise linear minimization

$$\min f(x) = \max_{1 \leq i \leq m} (a_i^T x + b_i)$$



$$\min_{x,t} t \quad \text{s.t.} \quad a_i^T x + b_i \leq t, \quad i = 1, \dots, m.$$

► Linear program with variables $x, t \in \mathbb{R}$.

LP Exercises

- 😊 Formulate $\min \|Ax - b\|_1$ as an LP ($\|x\|_1 = \sum_i |x_i|$)
- 😊 Formulate $\min \|Ax - b\|_\infty$ as an LP ($\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$)

Quadratic Programming

$$\min \quad \frac{1}{2}x^T Ax + b^T x + c \quad \text{s.t.} \quad Gx \leq h.$$

We assume $A \succeq 0$ (semidefinite).

Exercise: Say no constraints; does this QP always have a solution?

Nonnegative least squares (NNLS)

$$\min \quad \frac{1}{2}\|Ax - b\|^2 \quad \text{s.t.} \quad x \geq 0.$$

Exercise: Prove that NNLS always has a solution.

Primal problem

Let $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ ($0 \leq i \leq m$). Generic **nonlinear program**

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & x \in \{\text{dom } f_0 \cap \text{dom } f_1 \cdots \cap \text{dom } f_m\}. \end{aligned} \tag{P}$$

Def. Domain: The set $\mathcal{D} := \{\text{dom } f_0 \cap \text{dom } f_1 \cdots \cap \text{dom } f_m\}$

- ▶ We call (P) the **primal problem**
- ▶ The variable x is the **primal variable**
- ▶ We will attach to (P) a **dual problem**
- ▶ In our initial derivation: no restriction to convexity.

Lagrangian

To the primal problem, associate **Lagrangian** $\mathcal{L} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$,

$$\mathcal{L}(x, \lambda) := f_0(x) + \sum_{i=1}^m \lambda_i f_i(x).$$

- ♠ Variables $\lambda \in \mathbb{R}^m$ called **Lagrange multipliers**
- ♠ Suppose x is feasible, and $\lambda \geq 0$. Then, we get the lower-bound:

$$f_0(x) \geq \mathcal{L}(x, \lambda) \quad \forall x \in \mathcal{X}, \quad \lambda \in \mathbb{R}_+^m.$$

- ♠ Lagrangian helps write problem in **unconstrained form**

Lagrangian

Claim: Since, $f_0(x) \geq \mathcal{L}(x, \lambda) \quad \forall x \in \mathcal{X}, \quad \lambda \in \mathbb{R}_+^m$, primal optimal

$$p^* = \inf_{x \in \mathcal{X}} \sup_{\lambda \geq 0} \mathcal{L}(x, \lambda).$$

Proof:

- ♠ If x is not feasible, then some $f_i(x) > 0$
- ♠ In this case, inner sup is $+\infty$, so claim true by definition
- ♠ If x is feasible, each $f_i(x) \leq 0$, so $\sup_{\lambda} \sum_i \lambda_i f_i(x) = 0$

Lagrange dual function

Def. We define the **Lagrangian dual** as

$$g(\lambda) := \inf_x \mathcal{L}(x, \lambda).$$

Observations:

- ▶ g is pointwise inf of affine functions of λ
- ▶ Thus, g is concave; it may take value $-\infty$
- ▶ Recall: $f_0(x) \geq \mathcal{L}(x, \lambda) \quad \forall x \in \mathcal{X}$; thus
- ▶ $\forall x \in \mathcal{X}, \quad f_0(x) \geq \inf_{x'} \mathcal{L}(x', \lambda) = g(\lambda)$
- ▶ Now minimize over x on lhs, to obtain

$$\forall \lambda \in \mathbb{R}_+^m \quad p^* \geq g(\lambda).$$

Lagrange dual problem

$$\sup_{\lambda} g(\lambda) \quad \text{s.t. } \lambda \geq 0.$$

- ▶ **dual feasible:** if $\lambda \geq 0$ and $g(\lambda) > -\infty$
- ▶ **dual optimal:** λ^* if sup is achieved
- ▶ Lagrange dual is always concave, regardless of original

Weak duality

Def. Denote **dual optimal value** by d^* , i.e.,

$$d^* := \sup_{\lambda \geq 0} g(\lambda).$$

Theorem (Weak-duality): For problem (P), we have $p^* \geq d^*$.

Proof: We showed that for all $\lambda \in \mathbb{R}_+^m$, $p^* \geq g(\lambda)$.
Thus, it follows that $p^* \geq \sup g(\lambda) = d^*$.

Equality constraints

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_i(x) = 0, \quad i = 1, \dots, p. \end{aligned}$$

Exercise: Show that we get the Lagrangian dual

$$g : \mathbb{R}_+^m \times \mathbb{R}^p : (\lambda, \nu) \mapsto \inf_x \mathcal{L}(x, \lambda, \nu),$$

where the Lagrange variable ν corresponding to the equality constraints is unconstrained.

Hint: Represent $h_i(x) = 0$ as $h_i(x) \leq 0$ and $-h_i(x) \leq 0$.

Again, we see that $p^* \geq \sup_{\lambda \geq 0, \nu} g(\lambda, \nu) = d^*$

Duality gap

$$p^* - d^* \geq 0$$

Strong duality if duality gap is zero: $p^* = d^*$

Notice: both p^* and d^* may be $+\infty$

Several **sufficient** conditions known!

“Easy” necessary and sufficient conditions: **unknown**

Slater's sufficient conditions

$$\begin{array}{ll}\min & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad 1 \leq i \leq m, \\ & Ax = b.\end{array}$$

Constraint qualification: There exists $x \in \text{ri } \mathcal{D}$ s.t.

$$f_i(x) < 0, \quad Ax = b.$$

That is, there is a **strictly feasible** point.

Theorem Let the primal problem be convex. If there is a feasible point such that is strictly feasible for the non-affine constraints (and merely feasible for affine, linear ones), then strong duality holds. Moreover, in this case, the dual optimal is attained (i.e., $d^* > -\infty$).

Counterexample

$$\min_{x,y} e^{-x} \quad x^2/y \leq 0,$$

over the domain $\mathcal{D} = \{(x, y) \mid y > 0\}$.

Clearly, only feasible $x = 0$. So $p^* = 1$

$$\mathcal{L}(x, y, \lambda) = e^{-x} + \lambda x^2/y,$$

so dual function is

$$g(\lambda) = \inf_{x,y>0} e^{-x} + \lambda x^2/y = \begin{cases} 0 & \lambda \geq 0 \\ -\infty & \lambda < 0. \end{cases}$$

Dual problem

$$d^* = \max_{\lambda} 0 \quad \text{s.t. } \lambda \geq 0.$$

Thus, $d^* = 0$, and gap is $p^* - d^* = 1$.

Here, we had no strictly feasible solution.

Example

$$\begin{aligned} \min \quad & \sum_i x_i \log x_i \\ & Ax \leq b, \quad 1^T x = 1, \quad x > 0. \end{aligned}$$

Recall, convex conjugate of $f(x) = x \log x$ is $f^*(y) = e^{y-1}$.

$$\begin{aligned} \max_{\lambda, \nu} \quad & g(\lambda, \nu) = -b^T \lambda - \nu - \sum_{i=1}^n e^{-(A^T \lambda)_i - \nu - 1} \\ \text{s.t.} \quad & \lambda \geq 0. \end{aligned}$$

If there is $x > 0$ with $Ax \leq b$ and $1^T x = 1$, strong duality holds.

Exercise: Simplify above dual by optimizing out ν

Recap

- $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_i \lambda_i f_i(x) + \sum_i \nu_i h_i(x)$
- $g(\lambda, \nu) := \inf_x \mathcal{L}(x, \lambda, \nu)$
- $d^* := \sup g(\lambda, \nu) \leq p^* := \inf_x f(x) \quad \text{s.t. } x \in \mathcal{X}$ (weak duality)
- **Slater's** constraint qualification ensures $d^* = p^*$ (strong duality)

LP Duality

- ▶ Consider linear program

$$\min \quad c^T x \quad Ax \leq b.$$

- ▶ Corresponding dual is

$$\max \quad b^T \lambda \quad A^T \lambda + c = 0, \quad \lambda \geq 0.$$

- ▶ LP duality facts:

- If either p^* or d^* finite, then $p^* = d^*$, and both primal, dual problem have optimal solutions
- If $p^* = -\infty$, then $d^* = -\infty$ (follows from weak-duality)
- If $d^* = \infty$, then $p^* = \infty$ (again, weak-duality)

If LP is feasible, strong duality holds.
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Optimality conditions

$$\min_x f_0(x) \quad f_i(x) \leq 0, \quad i = 1, \dots, m.$$

- Recall: $\langle \nabla f_0(x^*), x - x^* \rangle \geq 0$ for all feasible $x \in \mathcal{X}$
- Can we simplify this using Lagrangian?
- $g(\lambda) = \inf_x \mathcal{L}(x, \lambda) := f_0(x) + \sum_i \lambda_i f_i(x)$

Assume strong duality and that both p^* and d^* attained!

Thus, there exists a pair (x^*, λ^*) such that

$$p^* = f_0(x^*) = d^* = g(\lambda^*) = \min_x \mathcal{L}(x, \lambda^*) \leq \mathcal{L}(x^*, \lambda^*) \leq f_0(x^*) = p^*$$

- Thus, equalities hold in above chain.

$$x^* \in \operatorname{argmin}_x \mathcal{L}(x, \lambda^*).$$

Optimality conditions

$$x^* \in \operatorname{argmin}_x \mathcal{L}(x, \lambda^*).$$

If $f_0, f_1, \dots, f_m$ are differentiable, this implies

$$\nabla_x \mathcal{L}(x, \lambda^*)|_{x=x^*} = \nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) = 0.$$

Moreover, since $\mathcal{L}(x^*, \lambda^*) = f_0(x^*)$, we also have

$$\sum_i \lambda_i^* f_i(x^*) = 0.$$

But $\lambda_i^* \geq 0$ and $f_i(x^*) \leq 0$, so **complementary slackness**

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m.$$

KKT Optimality conditions

Karush-Kuhn-Tucker Conditions (KKT)

$$f_i(x^*) \leq 0, \quad i = 1, \dots, m \quad (\text{primal feasibility})$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m \quad (\text{dual feasibility})$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m \quad (\text{compl. slackness})$$

$$\nabla_x \mathcal{L}(x, \lambda^*)|_{x=x^*} = 0 \quad (\text{Lagrangian stationarity})$$

- We showed: if strong duality holds, and (x^*, λ^*) exist, then KKT conditions are **necessary** for pair (x^*, λ^*) to be optimal
- If problem is convex, then KKT also **sufficient**

Exercise: Prove the above sufficiency of KKT. *Hint:* Use that $\mathcal{L}(x, \lambda^*)$ is convex, and conclude from KKT conditions that $g(\lambda^*) = f_0(x^*)$, so that (x^*, λ^*) optimal primal-dual pair.